Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim: Simulate dilation, erosion, opening and closing operation on images.**	
Experiment-11	**Date: 29-10-26**	**Enrollment No:92400133128**

AIM: Simulate dilation, erosion, opening and closing operation on images.

Theory:

Dilation, erosion, opening, and closing are morphological operations used in image processing to manipulate the shape and size of objects in an image.

Dilation:

Dilation expands the boundaries of objects in an image by adding pixels to the object's boundaries. It is achieved by sliding a structuring element over the image and assigning the maximum pixel value within the neighbourhood of each pixel.

Erosion:

Erosion shrinks the boundaries of objects in an image by removing pixels from the object's boundaries. It is achieved by sliding a structuring element over the image and assigning the minimum pixel value within the neighbourhood of each pixel.

Opening:

Opening is an erosion operation followed by a dilation operation. It is used to remove noise and small objects from the image while preserving the larger objects' shapes.

Closing:

Closing is a dilation operation followed by an erosion operation. It is used to close small gaps and holes within objects while preserving the overall object shapes.

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CODE:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread('Prac.jpeg', 0)

kernel = np.ones((5, 5), np.uint8)
# Perform Erosion
erosion = cv2.erode(image, kernel, iterations=1)
plt.imshow(erosion, cmap='gray')
# Threshold the image if it is not already binary
_, binary_img = cv2.threshold(image, 127, 255, cv2.THRESH_BINARY)

# Define a structuring element (e.g., 5x5 rectangle or ellipse)
kernel = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (5, 5))

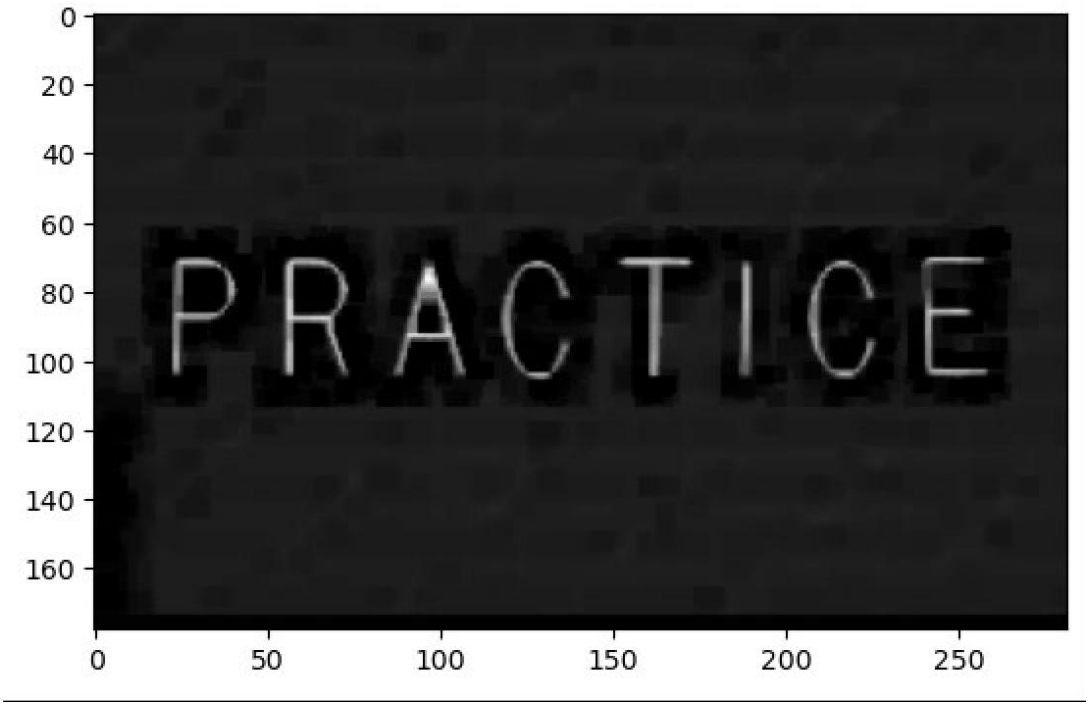
# Option 1: Dilation to bridge gaps directly
dilated_img = cv2.dilate(binary_img, kernel, iterations=10)

# Option 2: Closing (Dilation -> Erosion) to mend breaks and clean up edges
closed_img = cv2.morphologyEx(binary_img, cv2.MORPH_CLOSE, kernel)
```

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```
# Save or display results
cv2.imwrite('repaired_image.png', closed_img)
plt.figure(figsize=(10, 8))
plt.subplot(2, 3, 1), plt.imshow(image, cmap='gray'), plt.title('Original Image'), plt.axis('off')
plt.subplot(2, 3, 2), plt.imshow(erosion, cmap='gray'), plt.title('Erosion'), plt.axis('off')
plt.subplot(2, 3, 3), plt.imshow(dilation, cmap='gray'), plt.title('Dilation'), plt.axis('off')
plt.subplot(2, 3, 4), plt.imshow(opening, cmap='gray'), plt.title('Opening'), plt.axis('off')
plt.subplot(2, 3, 5), plt.imshow(closing, cmap='gray'), plt.title('Closing'), plt.axis('off')
plt.subplot(2, 3, 6), plt.imshow(closed_img, cmap='gray'), plt.title('Repaired Image'), plt.axis('off')
plt.tight_layout()
plt.show()
```

Original Image:



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Morphological image processing in cv2



Result:

The morphological operations—Erosion, Dilation, Opening, and Closing—were performed successfully on the grayscale image using a 5 × 5 structuring element. The repaired image was obtained by applying image processing techniques.

Conclusion:

Thus, morphological image processing operations were implemented successfully using OpenCV in Python to modify image structures and improve the appearance of the text.

GITHUB Link: https://github.com/abhinayreddy-7/DSIP.git